Zaraius Bilimoria

Robotics Engineering

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Education

Olin College of Engineering (B.S. Robotics Engineering) • Needham, MA • GPA: 3.97 — ***May 2028***

Skills

Programming & Development:	C++, C, Python, Java, MATLAB, Git, CMake, Bash, Tmux, Vim, Julia, Unity ML-Agents
Robotics & Embedded Systems:	ROS 2, Nav2, Docker, RViz, Foxglove, Gazebo, Ubuntu, Raspberry Pi, ArduPilot, STM32
Computer Vision and ML:	PyTorch, OpenCV, YOLO, TensorRT, ZED SDK, Open3D, DeepStream SDK
Design & Fabrication:	Onshape, SolidWorks, KiCad, 3D Printing, Soldering, Hand Tools, Power Tools

Experience

Robotics Intern • HITT Contracting • Falls Church, VA — ***Jun 2026 – Aug 2026***

- Architected a **ROS 2 social navigation system** for Boston Dynamics Spot, integrating human perception, prediction, and Nav2 motion planning for **operation around pedestrians** on active construction sites.
- Developed a C++ ROS 2 perception pipeline using **ZED 2i skeleton tracking** to estimate pedestrian position, velocity, and posture classification from 38 body keypoints, providing **real-time human state information for navigation**.
- Implemented a **custom Nav2 MPPI SocialCritic plugin** in C++ that predicts pedestrian trajectories and applies anisotropic Gaussian cost fields across 5,000 candidate trajectories per planning cycle, enabling **real-time socially-aware navigation** around stationary and moving pedestrians.
- **Integrated** perception, localization, and navigation into a **deployable ROS 2 autonomy stack**, debugging TF, sensor synchronization, and coordinate-frame issues on Boston Dynamics Spot.
- Built a Docker-based **TurtleBot3 simulation** environment for **rapid development** and repeatable validation of the autonomy stack before deployment to the physical robot.

Embedded Systems & Computer Vision Intern • Prophet AI • Somerville, MA — ***Jun 2025 – Dec 2025***

- **Designed and deployed a scalable 12-camera network** for a poultry farm using Raspberry Pis, replacing a legacy IP-camera system. Programmed H.264-over-RTSP stream in Python, ensuring 24/7 uptime with automated IP provisioning
- **Boosted inference throughput from 220 to 3,000 FPS** and **cut compute load by 75%** by training a lightweight PyTorch YOLO model and building a custom pixel-based motion filter with OpenCV, allowing continuous poultry health monitoring.
- **Increased the model F1 score from 0.94 to 0.99** by implementing a dynamic masking system to improve data quality, and by developing an **active-learning pipeline** which trains two CV models nightly, computes IoU disagreement, and flags discrepancies for retraining on flagged images.

Software & Integration Lead • Olin AERO • Needham, MA — ***Aug 2024 – Current***

- Taught core robotics software concepts (ROS 2, state machines, computer vision, and networking) to 6 first-year team members, structuring mini-lectures and projects to ensure contributions to the SUAS competition aircraft.
- Developed and led the implementation of the communication system for an autonomous aerial vehicle using **TCP** to reliably transmit live images and flight data from onboard **Pixhawk autopilot** and **Jetson Nano** to a ground station.
- Integrated **Pixhawk autopilot** for bidirectional control and data flow with onboard computing systems.

Robotics Software Engineer • Olin Agriculture Robotics Research Team • Needham, MA — ***Aug 2024 – Current***

- Designed and implemented **ROS 2 perception and sensor fusion** algorithms for real-time crop-row detection and global pose estimation, enabling autonomous navigation and alignment with crop rows in unstructured environments.
- Implemented **3D Gaussian splatting techniques** to process depth camera data, generating dense **3D point clouds** for weed identification and path planning.
- Designed schematic and fabricated a protoboard-soldered signal-conditioning interface for commercial resistive soil probe.

Captain, Lead Programmer; NUTRONs 125 • FIRST Robotics • Revere, MA — ***Jan 2019 – Jul 2024***

- Led a 19-member team to create a robot ranked 2nd out of 600+ teams at the global championship.
- Collaborated with professional mentors and mechanical team to develop autonomous & teleoperated robot code in Java.
- Mentored junior programmers through code reviews, debugging, and subsystem development, improving team capability.